

11L2-svd-pca-recommender

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Slide 1

Content

Description

This slide is the title slide for a lecture on Artificial Intelligence Basics. It displays the course title, "AI Basics," the course number, "Purdue CS24300," and the instructor's name, "Tony Bergstrom." It also includes a credit line, "Slides adapted from Jen Neville, CS 24200," indicating the source of the lecture materials. The overall appearance is a simple, dark gray background with light gray text, typical of a presentation slide. The context of the course is clear: this is the introductory slide to a lecture on Artificial Intelligence.

Figures

Slide 2

Content

Matrix Factorization

Description

This slide, likely the title slide for a section on Matrix Factorization, displays the single heading "Matrix Factorization" in a simple, sans-serif font. The background is a neutral gray, typical of presentation slides. The context of the course (Artificial Intelligence Basics) suggests that this is a topic within a larger discussion of techniques or algorithms related to machine learning or data analysis, where matrix factorization is a key component.

Figures

Slide 3

Content

Matrix Factorization Motivation

High-dimensional data will often have simpler structure, e.g.

- Topics in documents/products
- Product characteristics (reliability, cost etc.)
- User preferences/demographics
- Etc.

These effects can be approximated using a lower-dimensional model

Description

This slide, titled "Matrix Factorization Motivation," presents a rationale for using matrix factorization techniques. It argues that high-dimensional data, such as data related to topics in documents or products, product characteristics (like reliability and cost), and user preferences/demographics, often exhibit underlying simpler structures. The bullet points list examples of such structures. The final sentence states that these underlying structures can be approximated using a lower-dimensional model, implying that matrix factorization can be used to reduce the dimensionality of the data while preserving the essential information. The context of the course (Artificial Intelligence Basics) suggests that this slide is part of a larger discussion on data analysis and dimensionality reduction techniques within the field of machine learning.

Figures

Slide 4

Content

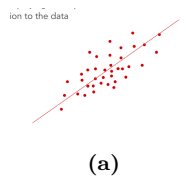
Matrix Factorization Motivation

Usually, the data vary more strongly in some dimensions than others. Fitting a linear (simplifying assumption) model to the (k) strongest directions gives a good approximation to the data.

Description

This slide, titled "Matrix Factorization Motivation," discusses the rationale behind matrix factorization techniques. It states that data often exhibit stronger variations in certain dimensions compared to others. The slide visually supports this claim with a scatter plot. A red line is drawn through the data points, representing a linear model. The text further explains that fitting a linear model (a simplifying assumption) to the k strongest directions (implied by the scatter plot) provides a good approximation of the data. The context of the course (Artificial Intelligence Basics) suggests that this slide is part of a lecture on dimensionality reduction techniques, where matrix factorization is presented as a method to capture the most significant variations in the data.

Figures



Slide 5

Content

Matrix Factorization

Columns of B represent a vector of latent "factor" values for the n^{th} movie

Rows of A represent a vector of "factor" values for the m^{th} user

Choice of k determines how many latent factors to use

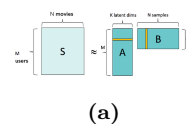
Embeddings: vector of latent factor values (for an item) projects it down into k -dimensional space

[fill=lightgray!20] (0,0) rectangle (2,2); at (1,1) S ; at (0.5,2.5) N movies; at (-0.5,0) M users; [fill=lightgray!20] (3,0) rectangle (5,2); at (4,1) A ; at (3.5,2.5) K latent dims; [fill=lightgray!20] (6,0) rectangle (8,2); at (7,1) B ; at (6.5,2.5) N samples; at (4,-0.5) M ; at (7,-0.5) K ; [thick, yellow] (3.5,1) – (4.5,1); [thick, yellow] (6.5,1) – (7.5,1);

Description

This slide, titled "Matrix Factorization," describes the concept. It explains that in matrix factorization, columns of matrix B represent a vector of latent "factor" values for each movie (n^{th} movie). Rows of matrix A represent a vector of "factor" values for each user (m^{th} user). The choice of k determines the number of latent factors to use. The slide also introduces the concept of "embeddings," which are vectors of latent factor values that project an item (movie or user) down into a k -dimensional space. Visually, the slide depicts three matrices (S , A , and B) with their dimensions labeled. Matrix S represents the movies, matrix A represents the users, and matrix B represents the samples. The dimensions of each matrix are labeled (e.g., N movies, M users, K latent dimensions, N samples). Yellow lines highlight the shared dimension (M) between matrices S and A , and the shared dimension (K) between matrices A and B . The context of the course (Artificial Intelligence Basics) suggests that this slide is part of a lecture on recommendation systems or collaborative filtering, where matrix factorization is a key technique for finding latent factors that describe users and items.

Figures



Slide 6

Content

Two Dimensional Embedding of Movies

[xlabel=Factor vector 1, ylabel=Factor vector 2, xmin=-1.5, xmax=1.0, ymin=-1.5, ymax=1.5, xtick=-1.5,-1.0,-0.5,0.0,0.5,1.0, ytick=-1.5,-1.0,-0.5,0.0,0.5,1.0,1.5, title=Two Dimensional Embedding of Movies, legend pos=south east, legend style=font=,] [only marks, mark=*, mark size=2pt, color=blue] coordinates (-1.2, 1.2); Freddy Got Fingered; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-0.8, 1.0); Freddy vs. Jason; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-0.5, 0.8); Road Trip; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-0.9, 0.5); Half Baked; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-1.0, -0.5); Natural Born Killers; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-0.7, -0.7); Kill Bill Vol. 1; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-0.5, -1.0); Julien Donkey-Boy; [only marks, mark=*, mark size=2pt, color=blue] coordinates (-0.2, -1.2); Scarface; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.2, -0.8); The Wizard of Oz; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.5, -0.5); The Longest Yard; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.7, -0.2); The Fast and the Furious; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.9, 0.0); Armageddon; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.6, 0.7); Catwoman; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.4, 1.0); Coyote Ugly; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.8, 1.2); Punch-Drunk Love; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.7, 0.9); The Royal Tenenbaums; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.5, 0.7); Being John Malkovich; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.3, 0.5); Belle de Jour; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.1, 0.3); Citizen Kane; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.9, -0.9); Annie Hall; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.7, -1.2); Sophie's Choice; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.5, -1.4); Moonstruck; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.3, -1.6); The Way We Were; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.1, -1.8); The Waltons: Season 1; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.9, -1.0); Maid in Manhattan; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.7, -0.7); Lost in Translation; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.5, -0.5); Stepmom; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.3, -0.3); Sister Act; [only marks, mark=*, mark size=2pt, color=blue] coordinates (0.1, -0.1); The Sound of Music;